useless. The end of the equalizers next the broken spring or hanger will be found pressing hard against the frame, and it should be drawn away from the frame sufficiently to put a nut or other suitable piece between end of equalizer and frame. A chain and jack, or by suitable bar leverage the equalizer may be sufficiently eased to get nut or block in place. An engine so blocked must be run carefully, as the driving box is very liable to heat, and must be well and constantly oiled.

78—With broken equalizer?

A.—If the equalizer key has broken it is possible to clamp the equalizer so as to hold it approximately in place. If the equalizer is broken entirely, the free ends of the driving springs must be clamped down and chained to the frames, or blocks be put on top of the boxes and under frame. Care must be taken in any case to see that lubrication is given freely.

79—With broken engine truck spring or hanger?

A.—Jack up engine on disabled side and block on top of engine truck box and below engine truck frame. If spring is broken badly, block on top of both boxes on disabled side. If the hanger is broken, block on top of box next to it and chain free end of spring up to the engine truck frame. Look well to lubrication.

80—With broken engine truck centre pin on Mogul, what is to be done?

A.—The breaking of the engine truck centre pin on a Mogul can be temporarily repaired by jacking up the engine, removing the broken pin and chaining up the free end of the equalizer by a chain passing up through the hollow centre casting and secured by a bolt placed horizontally on top of the centre casting. Another method would be to block the equalizer down under the cylinder saddle as near to the cross equalizer as possible. The engine must be run most carefully under these circumstances, as the guiding influence of the engine truck has practically disappeared.

81—With broken intermediate equalizer on Mogul?

A.—The broken intermediate equalizer lets down the frame on the main and the back driving boxes on one side. Block on top of both boxes, as both driving and trailing driving springs are useless. Proceed with care, as both boxes are liable to heat.

82—What do you do when a tire breaks and comes off the wheel on standard engine?

A.—Take down the cellar and put a block of wood in the cellar, cover with well oiled packing, then block up the box of the disabled wheel by inserting a nut or other suitable piece on top of the pedestal binder and under the box. This will carry the weight of the wheel clear of the rail. Disconnect side rods and main rod at butt end, push rod, crosshead, etc., forward, block crosshead, disconnect valve rod, centre valve and clamp, shut off lubricator and proceed very cautiously.

83—With front tire on Mogul or ten-wheel engine?

A.—Proceed as outlined in answer to question No. 82, as far as the blocking of the box is concerned, and in addition disconnect front section of side rod on both sides and run cautiously, as the disabled wheel is no longer able to help in guiding the engine round track curving toward the other side of the engine.

84—Main tire on Mogul?

A.—Proceed as outlined in answer to question No. 82.

85—With the back tire on Mogul?

A.—Proceed as outlined in answer to question No. 82, and in addition take down back section of side rod on both sides.

86—With both back tires on Mogul?

A.—Block up both back wheels, disconnect back section of side rods on both sides; but it is not necessary in this case to put block in cellar, as these wheels will not revolve. The main drivers will have to be carefully oiled, as they will carry more weight than usual. Where the overhand is long it is necessary to carry some of the weight of the engine on the tender, and this may be done by chaining two pieces of rail to the back portions of the engine frame, one rail on each side. Each rail chained in two places, so as to make an extemporized extension of the frames, and jack the engine high behind and chain from the end of one projecting rail over end sill of tender and down to other projecting rail, lower engine and part of the weight of the engine will be transferred to front of tender. The disabled wheels will ride clear of the rails. Look after the oiling of the journals on the tender on account of the increased weight so carried.

87—With back tire or back driver broken off, how do you fix engine so you can back around curves when necessary?

A.—Proceed as outlined in answer to question No. 86, but back very slowly and cautiously round curves with engine in this condition.

88—At what points is weight of engine carried when blocked up over the back driving box?

A.—The weight is carried directly by the frame over the back box and by equalizer pivot, and by the other springs and hangers in the usual way.

89—If driving box or brass breaks so it is cutting the axle badly, what can you do to relieve it?

A.—Take down the driving box cellar and put block of wood in the cellar and cover with well oiled packing. Run wheel up on wedge, or block so as to carry the box up in the jaws, put nuts or other suitable pieces on top of pedestal binder and below box. When the wheel comes on the rail again part of the weight will be taken off the defective brass; oil thoroughly and often and proceed carefully.

90—Do you consider it an engineer's duty to have suitable hardwood blocks on his engine to use in case of a breakdown?

A.—Yes.

Elements of Physical Science.

IV.—GRAVITY.

The tendency of bodies when unsupported is to fall to the ground. This is called gravity, and this law of attraction is universal. It is not confined to the surface of the earth alone, but extends through space, and, as has been already stated, is the great agent by which the stars are kept in their spheres. This law of gravitation is a constant principle and acts instantly, and is not lessened by the interposition of any other substance, and is also entirely independent of the nature of matter. It has been scientifically demonstrated that the action of the sun is found to be the same on all of the heavenly bodies. It is this force of attraction or gravitation that keeps all movable bodies on the surface of the earth from falling into space.

It may be noted that the force of gravity increases as the amount of matter increases, and the force of gravity decreases as the square of the distance increases. In regard to the planetary system of which the earth forms a part, the sun is 800 times greater than all the planets put together. It is on this account that its attraction is felt by the remotest bodies of the solar system. Weight of bodies is simply a measure of the force with which bodies are drawn towards the earth, and if the earth contained twice the matter which it now contains all bodies movable on its surface would be found to have twice their present weight. As gravity decreases as the square of the distance from the earth's centre increases, it follows that bodies become lighter in proportion to the distance they are removed from the earth's surface. A mass of iron weighing one ton on the earth's surface would be found at the distance of 4,000 miles to weigh only a quarter of a ton. The varying weight on the earth's surface is small. Four miles above the surface of the earth a body weighing 1,000 pounds becomes only two pounds lighter.

FALLING BODIES.

All bodies acted on solely by gravity fall to the earth with the same velocity. It is true that a metal ball will fall faster than a feather, but this is not owing to a variation in the fixed law of gravitation, but arises from the effect of the atmos-

information for myself, I would thank you to explain why the brakes failed to stop the front portion of train. I will explain the last case I handled in detail.

A freight train, consisting of 50 loaded coal cars, 3,500 tons. Engine had 9½ inch air pump. Main reservoir, 65,000 cubic ins. All Westinghouse equipment, including caboose, which was piped.

While descending a grade (70 ft. to the mile) at a speed of 18 or 20 miles per hour, the engineer attempted to make a service application of the brakes, when all brakes worked quick action causing such a shock to the rear of the train as to detach the caboose from train, knocking the conductor out of the cupola and slightly injuring him. When he had recovered from the shock he discovered that the caboose was detached from the train and was following the train some three or four car lengths from the rear car. He let the caboose follow and recoupled to train about ½ of a mile from the point at which it became detached. After coupling the caboose he closed the angle cock on the rear car and proceeded on down the grade until stopped by an automatic block signal, where the engineer made the stop with a service application of the brakes. At this point, a defective triple, which caused the quick action, was located and cut out. At the investigation the engineer stated that when he attempted to apply the brakes at the point where the caboose broke off that his brake valve service exhaust closed, and having had trouble with a defective triple in this train he immediately placed the brake valve handle in release position to keep the train from stopping, and that he knew nothing of the train line being open on the rear of train until told by the conductor.

I will be glad to have you explain through the columns of your valuable paper why the brakes failed to stop this train with the angle cock open on the rear.

C. M. KIDD,
Gen'l Air Brake Inspector.

Roanoke, Virginia.

Driver Brakes and Other Matters.

Editor:

The driver brakes on locomotives, both in freight and passenger service, are receiving more attention at present than they have been in the past, yet there is considerable room for improvement. A careful inspection of their condition, and to note the number of wrongly used triple valves and auxiliary reservoirs, will convince an air brake man that the driver brake is not being maintained as it should be.

Some air brake repairmen seem to be utterly unable to distinguish the difference between the F 46 and H 24 triple valves, in spite of the fact that the catalogue plate number is cast in raised letters on the body of both valves; therefore, it is not surprising that they cannot remember the size of the auxiliary reservoir, which is always 2 ins. larger than the brake cylinder in diameter.

However, this wrongly used equipment is mostly due to the lack of material with which to make repairs, and the repairmen are compelled to replace broken or missing parts with something, regardless of size or style.

Probably 50 per cent. of the engines running in freight service have a defective driver brake. In many cases the engineer or inspector will test the brake cylinders and pipe connections with a torch, note that the piston travel is correct and decide that the brake is all right, where, if an air gauge were attached to the cylinder, it would show that the hand will fall from fifty to ten or fifteen lbs. in a very few seconds and remain there for a considerable length of time, showing no leakage.

The importance of a good driver brake has been dwelt on in these columns at various times. It has been shown that the braked weight of a heavy freight engine is equal to that of about five large capacity freight cars.

An inferior driver brake is not only annoying in heavy grade work, but it is dangerous, and the damage done to a train amounts to many times the cost of its proper maintenance. The intense heat of a locomotive has a tendency to dry out the packing leathers and joints and start leaks. The jar and pound of the engine will start pipe joints to leaking. The leaks are stopped from time to time, but will start again, and where an auxiliary reservoir is used to store a supply of compressed air for the operation of the brake, the leaks quickly destroy the braking power.

The distributing valve, when used, overcomes this trouble entirely, although it should not be considered an excuse for leaky pipe joints and packing leathers. This valve can be applied to an engine in a very short time, the main reservoir, brake pipe and brake cylinder connections made, the connection to the application chamber plugged, and can be operated with the G 6 brake valve.

The tender brake can be operated from the distributing valve by the addition of one piece of pipe and a hose, and if it is desired to leave the triple valve and auxiliary reservoir on the tender to keep it interchangeable it can be done by the addition of the straight air brake double-seated check valve placed between the triple valve and brake cylinder, with the pipe from the distributing valve connected to the straight air side, and the triple valve can be operated in conjunction with the distributing valve or cut-out.

Of course, the entire E. T. equipment should be used, and it is more economical in every way.

At a conservative estimate, and due principally to hard handling, the G 6 brake valve will require cleaning four or five times to once for the H 5 brake valve, and owing to improvement in the construction of the H 5 brake valve, its rotary valve and seat will require facing off and grinding once, while a G 6 valve, in the same class of service, will be ground three or four times.

The economy in the use of the H 5 valve is apparent, as one-eighth of an inch removed from the wearing surface of the valve and seat in the G 6 valve is all that is permissible.

Under the tender there is no comparison between the two brakes, as it is all in favor of the E. T. brake.

There is no cost of cleaning and repairs to triple valves, and this is quite an item, as the tender triple catches most of the dirt and scale in the brake pipe and more or less moisture which is thrown into the brake cylinder during emergency applications.

Above all, there are no detentions due to defective triple valve or broken cross-over pipe, and no auxiliary reservoir to be eaten up by rust, which requires about half a day's time to renew.

The work of the air pump is more evenly distributed. During the application of the brake the air pump restores in the main reservoir the air that is expanded into the driver and tender brake cylinders, and in the release the volume from the main reservoir is all used to release and recharge the train, instead of a considerable portion being used to recharge two large auxiliary reservoirs on the engine and tender, the combined capacity of which is equal to about seven of the eight in freight auxiliaries.

This feature allows the pump more time in which to compress a given amount of air, and in a measure reduces the liability of the pump becoming overheated, and it is a well known fact that an overheated air pump is the worst disorder that the air brake can have. It not only reduces the capacity of the pump, but excessive heat ruins the air valve packing rings and air cylinders and scatters hot and burning oil throughout the entire air brake system, and in a few days all the different valves are gummed up and out of order.

The E. T. equipment and quick service triple valves should double the length of time the pump can run without being overhauled and consequently reduce the cost of repairs at least 50 per cent. Where four men are employed to keep up the repairs to the standard Westinghouse brake, two men should be able to do it with the E. T. equipment.

G. W. KIEHM.

Washington, D. C.

traverse of the steam through cavities lying outside of the influence of heat passing through the smokebox.

It would seem that the Walschaerts valve gear on the engine tested had been designed more nearly correct in principle than has been usual, to make so fair a showing as to tonnage hauled as this machine has, in the face of the disclosed facts. I wonder in how many other cases this valve gear has been condemned on account of faulty design elsewhere. The indicator cards taken during these tests will, of course, be interesting, and no doubt will form the subject of a discourse by some one; but what do the cards indicate? Simply the personal theory of the designer of the valve motion in regard to lap, lead and exhaust clearance.

I wish to submit the following propositions to those who are interested in the matter, and who have a fair knowledge of the construction of both the Walschaerts and Stephenson valve gears; it is a fact that different engine designers and motive-power officials have differing beliefs as to what should be the proper proportions of the valve for an engine in any certain class of service; now:

First—If the designer's ideas are incorporated in a valve as applied to a locomotive equipped with the Stephenson link motion whereby the admission of steam to the cylinder begins when the crank-pin is at a predetermined point in its revolution and is cut off at another given point, and the exhaust begins when the crank-pin has reached a calculated position, can not this engine be fitted with the Walschaerts gear and have its valve so designed, and the combination lever so proportioned that the valve's phases will occur identically the same as they did with the Stephenson gear? May not the opening and closing moments be coincidental with both gears? It is a matter wholly determined by the amount of steam lap and exhaust clearance of the valve and the proportions of the combination lever; by working on those features a valve theorist can make the Walschaerts gear give his own pet steam distribution in the cylinders.

Second—If two engines, with the Stephenson and Walschaerts motions, respectively, but of the same general design in all other respects and equally proportioned, can have their valves and valve gear so devised that the steam distribution to the cylinders is the same in both—the opening and closing moments occurring when the crank-pin is at the same identical points on both engines—can there be any reason why one engine should develop a greater tractive power than the other?

When a service change in the cut-off is made by hooking up the engine, then is the time when the only automatic difference between the two valve gears comes in, in so far as the results obtained are concerned; whatever lead the Walschaerts valve may have had in full gear remains the same when the reverse lever is hooked up—it becomes no greater; while with the Stephenson gear the lead does increase as the increased speed compels hooking up. Instead of this increase of lead being a help to a fast-running engine, as some declare, it may be a detriment, for it can hardly be disputed that most heavy passenger engines with the Walschaerts gear are superior to the Stephenson-geared engines of the same class at high speeds, and the lead of the Walschaerts engines is non-increasing.

Third—With reverse lever in any given notch the steam admission port to cylinder is open while the crank-pin is describing a certain number of degrees of the wheel's circle. Increasing the amount of lead, or decreasing or removing it entirely, will not cause the steam to be admitted to the cylinder for any longer or shorter proportion of the crank-pin travel; the crank will



NEW FRENCH ENGINE AND TRAIN.

receive the steam impulse during the same number of degrees of its revolution, but its zone of power may be shifted and that is all. Lead cannot increase the power.

Fourth—When the opening and closing of the admission ports occur with the crank-pin an equal number of degrees each side from the quarter points (which coincide with a vertical line through the center of the wheel hub) the axle is then receiving the greatest turning force that the piston can give it—the leverage position is true; this correct placing of the degrees of force is impossible when lead exists, with any type of valve gear. Lead thus decreases the power of the crank.

Fifth—With lead opening of the admission port when the crank-pin is on the dead center a certain number of the crank's degrees of force *are annulled completely* as to their turning power, but then exist as a *retarding power*, giving a braking effect between axle and driving-box. *Lead is thus destructive of power.*

Now, for experimental purposes I should like to see a locomotive valve gear designed without any lead at all—with the valve always one-fourth of a cycle of motion from the piston. That would not be possible with the Stephenson double-eccentric motion, in which it was early learned that the eccentrics would have to be shifted away from their positions normally quartering with the crank-pin, but it would be easily possible with so simple and square a gear as the Walschaerts, which would only require the removal of the combination lever, coupling the radius rod directly to the valve stem, although this would make it necessary to lengthen or shorten the throw of the eccentric, according as to whether the valve was of outside or inside admission.

The design of *the valve* would then demand most careful consideration; but as the question of the correct delivery by the Walschaerts gear lies wholly within the valve and the doubtful combination lever, and remembering that with but one eccentric, and it exactly ninety degrees from the crank-pin, the valve has a most rapid travel while the crank is near to, and passing, the dead-center points, and that when the engine on one side is on the dead center the engine on the other side is very much alive. It might be well to go back to first principles with the Walschaerts gear and try it out by throwing the whole government of the steam distribution on the valve itself, without any interference from eccentric or crosshead.

There is a strong probability that not one man now alive who is interested in locomotive valve motion has ever witnessed the action of double-connected engines, such as a locomotive, with valves set—and working squarely at all points of cut-off—at exactly ninety degrees of the cycle of motion from the piston and, for reasons stated, it has not formerly been clearly possible. But by using only that portion of the Walschaerts gear that would reduce to first principles the action of the valve, and by substituting different valves with differing combinations in the amount of steam lap and exhaust clearance, some indicator cards—and incidentally some unlooked-for work on the part of the engine—may be expected.

W. W. Wood,
Air Brake Instructor, C. I. & L. R'y.
Lafayette, Ind.

Mr. Thomas Purvis, who was not long ago appointed superintendent of motive power of the Denver and Rio Grande, is the Purvis mentioned in Skeever's Object-Lessons, and he it was who is responsible for many of the points in connection with shop work which were introduced into those inimitable sketches of "Skinny's" life and career on the Great Air Line.

The cut-out cock for the H 6 valve is the Penna. R. R. standard cut-out cock; the brake pipe exhaust port of the brake valve is piped to the small end of this cock and a connection to the brake pipe is made from the large end.

The supply of air for the brake pipe feed valve is taken from a point between this cock and the brake valve, so that when this cock is closed it cuts off all pressure from the brake valve and opens a communication between the brake pipe, the feed valve pipe, and the valve body of the brake valve, and at the same time closes the brake pipe service exhaust port. When double-heading the cut-out cock under the H 6 brake valve should be closed on the second engine and both brake valve handles placed in their running positions. As the brake valve is still connected with the brake pipe, the gauges will show at all times the pressure in the brake pipe, and the brake on the engine and train can be applied by moving the valve handle to emergency position.

The independent brake can be applied or released at any time regardless of the position of the equalizing valve.

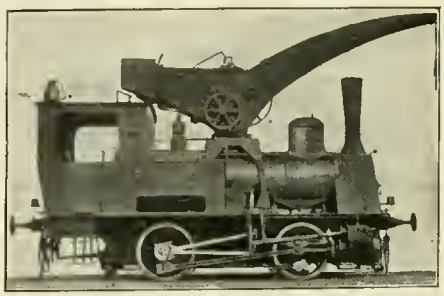
With the brake valve handles in their running positions the exhaust port of the application chamber of the distributing valve is open to the atmosphere by the way of the independent brake valve and the emergency exhaust port of the H 6 brake valve. When the S 6 brake valve handle is placed on lap position all ports are blanked, and when placed in application position, air from the reducing valve will flow into the application chamber of the distributing valve, and, if the equalizing valve is in release position, it will pass through its exhaust port and back as far as the S 6 rotary valve seat. When the handle is returned to running position, the pressure will escape through the exhaust port of the H 6 valve.

If the equalizing valve is in application position when the S 6 valve is used, it will remain applied when the handle is again placed in running position, until the brake pipe pressure forces the equalizing valve to release position or the independent valve handle is placed in release position, when the application chamber will be emptied through the independent brake valve exhaust port.

From this it will be seen that the brake will be applied with both brake valve handles in their running positions, if the brake pipe is overcharged and a few small leaks exist, consequently there will be some complaints of brakes "creeping on" from those who handle the brake without regard for the proper positions of the valve handles. The H 5 brake will not apply under these conditions, as the application chamber would be open directly to the atmosphere and the flow of air from the pressure chamber would be free to escape.

The pipe which connects the train line exhaust to the cut-out cock under the H 6 valve should be tested for leaks occasionally by closing the cock and making a service reduction; the object of this arrangement is to prevent a loss of brake pipe pressure when double-heading.

The rapid increase of brake pipe pressure during train brake release is likely to force the brake valve equalizing piston of the second engine from its seat, and it would be a difficult matter to get the piston to seat without opening the cock and admitting main reservoir pressure to the brake valve. The pipe connected to the large end of the cock admits brake pipe pressure, which, assisted by spring pressure, holds the plug valve to its seat. Should this pipe on the leading engine break off, the brake pipe leak can be plugged and a blind washer inserted in the union connection at the cock; should main reservoir pressure force the valve from its seat, the pressure will equalize instantly and the spring again seat the valve.



A KRUPP CRANE TANK ENGINE.

Should the pipe break off on the second engine, the brake pipe leak can be stopped and the handle on the cock turned to a position half way between open and closed, and a wedge driven between the handle and the body to hold valve to its seat. In this position all three ports in the cock will be blanked, and as the valve handle is in running position, air will flow through the ports in the brake valve to the feed valve pipe and feed valve, unseat the supply valve and flow through the reservoir pipe to the top of the rotary valve.

Under ordinary conditions when the H 6 brake valve is placed in release position air flows from the main reservoir, to the brake pipe and equalizing reservoir, from the main reservoir to the low pressure governor top and from the feed valve pipe through the warning port to the atmosphere. If an application has been made previously, air will flow from the distributing valve as far as the rotary valve seat. In running and holding positions, air flows from the feed valve pipe to the brake pipe from the main reservoir to the low pressure governor top and from the brake pipe to the equalizing reservoir.

When in running position, after the brake has been applied, air will flow from the distributing valve to the atmosphere. On lap position all ports are blanked and when placed in service position, the equalizing discharge valve and reservoir control the flow of brake pipe pressure, same as all other Westinghouse brake valves.

In emergency position air flows from the brake pipe to the atmosphere, from the equalizing reservoir to the atmosphere, and from the feed valve pipe to the application chamber of the distributing valve through the non-return check valve.

The distributing valve has the same stop cock arrangement for cutting out tender driver brake and truck brake cylinders. The stop cock for cutting out the distributing valve is located in the supply pipe. The pipe connecting the brake valve with the distributing valve is used for the same purpose that the retaining valve pipe is used with the ordinary triple valve, therefore it should not be plugged if accidentally broken.

If the application chamber pipe is broken the application chamber can be plugged and the independent brake cannot be applied.

Washington, D. C. G. W. KIEHM.

Erosive Action of Steam.

Not long ago the Hon. Charles A. Parsons, at a meeting of the Royal Institution of Great Britain, showed an enlarged photograph of a hardened steel file of square section, which for 145 hours had been exposed to the action of a jet of steam at 100 pounds pressure discharged into a condenser with a pressure of one inch, absolute, of mercury. The cutting action of the jet upon this extremely hard material was very remarkable. The file was smoothed down where the steam had passed over it, while the teeth on the other parts of the file were as they had come from the manufacturer.

In speaking of the erosive action of steam Mr. Parsons said, "It has been shown by experiment that if drops of pure water arising from the condensation of expanding steam impinge on brass at a greater velocity than about 500 feet per second, there results a slow wearing away of the metal. It is very slow and would require about ten years to erode the surface to a depth of one thirty-second of an inch. In the compound turbine the striking velocity is much below this figure."

One of the pioneer railroads constructed in New Jersey was the Jersey City to Paterson now used by the Erie. When the chief engineer engaged in the survey of this railroad made his first report to the directors, he informed his employers that he had introduced several reverse curves in the location for the sake of their graceful and picturesque effect.